

A Guide for Using SeisSol to Study Small Solar-System Bodies

From: PsycheESE: A Simulation-Based Investigation of Using Impact Seismology to Study Asteroid Interiors
Spencer Emmons¹, Elysian Hudson¹, Caleb Zick¹, Jacob Coons¹, Daniel Joseph¹, Syed Hamza Ayaz¹

¹School of Earth and Space Exploration, Arizona State University, Tempe, AZ



The open-source software SeisSol (Gabriel et al., 2025) is a powerful seismic simulation tool that utilizes the arbitrary high-order accurate derivative discontinuous Galerkin method (ADER-DG) to simulate wave propagation. It also utilizes meshes to approximate three-dimensional model geometries and specifies material elasticity to approximate geological subsurface properties. This guide intends to provide additional help for researchers who want to use SeisSol to study the seismic effects of impacts on small bodies in our solar system, such as asteroids like (16) Psyche. Much of the process outlined here is identical to using SeisSol for simulations of quakes on Earth, the Moon, or Mars. As such, this guide primarily focuses on the aspects that are unique to our use case, including possible pitfalls for how this software works on small bodies as well as how to avoid them. Note: the version of SeisSol this guide was written for was built on March 16, 2026, and newer versions may require adjustment to this process.

1. Install software following the steps in SeisSol's documentation, here: <https://seissol.org/>
2. Prepare the various input and parameters files. Our recommendation would be to copy the example WP2_LOH1 files from the SeisSol github and edit them to suit your needs, though you can write them from scratch if you prefer. The files breakdown as follows:
 - a. Parameters: parameters.par
 - i. You can use this to define the filenames of the other inputs and control the timestep, simulation length, and simulation time between saves.
 - b. Material Characteristics: loh1.yaml
 - i. Each group has 3 variables, rho is the density, mu is the shear modulus, and lambda is Lamé's first parameter. For these, make sure to use units of kg/m^3 for the density, and Pascals for the others.

- c. Source: LOH1_source_50.dat
 - i. You do not edit this file directly, instead, create it by editing and running the `generate_LOH_source_type50.py` Python script. Note: The authors used the older point source version (type 50) as it is better documented, the other sections of this guide should apply to either this, or the newer type 42. However, this section about generating the source data will not be as useful.
 - ii. Best practice is to ensure the source is entirely within your model (but very near the surface for an impact). Otherwise, much of the energy will be lost to space instead of creating seismic waves. This can be done through manipulation of the source's x, y, and z coordinates and the source's size.
 - iii. To edit the source time, change the second number in the "vtime" variable of the Python script. You will also need to adjust the variable "T" to ensure the integral of the moment rate is still equal to 1. Note: since the end of the header are the dt and total sample points, remember to update these along with the numbers in the moment rate function.
 - 1. As impacts are typically very fast, the source time will also be short which will lead to a smaller source dt. If this gets smaller than the simulation dt, it can cause issues — the authors did not experience this in our use, but we were warned about the possibility by one of SeisSol's developers.
 - d. Mesh and Receivers
 - i. These are mentioned together as their possible pitfalls in this use case are largely intertwined.
 - ii. The `recordPoints.dat` file defines the locations of the receivers, or simulated seismometers. It is simply a list of the x, y, and z coordinates of the receivers. Each line defines a new receiver. IMPORTANT: Coordinates in this file only accept positive values, leading into the next point...
 - iii. It is recommended that you place the model such that all coordinates are zero or positive. While centering it on the origin may be simpler in terms of making the model, it will cause issues should you wish to place receivers in the "negative" parts of the model.
- 3. Run the simulation.
 - 4. Read and visualize the output.

This work was created in partial fulfillment of Arizona State University Capstone Course "SES 410/411". The work is a result of the Psyche Student Collaborations component of NASA's Psyche Mission (<https://psyche.ssl.berkeley.edu>). "Psyche: A Journey to a Metal World" [Contract number NNM16AA09C] is part of the NASA Discovery Program mission to solar system targets. Trade names and trademarks of ASU and NASA are used in this work for identification only. Their usage does not constitute an official endorsement, either expressed or implied, by Arizona State University or National Aeronautics and Space Administration. The content is solely the responsibility of the authors and does not necessarily represent the official views of ASU or NASA.